Sahaj Bhadja

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Summary

Web software engineer with 2+ years shipping **React and TypeScript learning applications**, real-time collaborative experiences, and AI study workflows; experienced in browser and network behavior, async systems, performance optimization, testing, code review, and iterative delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built and evolved **user-focused learning platform services** for 500+ users, optimizing indexed access and async browser-facing workflows to reduce dashboard latency from 1.2 seconds to 350 milliseconds while maintaining secure isolation.
- Implemented reliable async application behavior across 5,000+ monthly workflows, using explicit state transitions, failure isolation, validation, and reconciliation while collaborating with stakeholders to ship tested enhancements through CI/CD with 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Designed and shipped **React and TypeScript learning experiences** for uploads, question management, grading history, and reporting, integrating Flask APIs that served 150+ instructors and handled 10,000+ daily requests.
- Integrated OpenAI, Claude, and Qwen models behind switchable APIs, critically validating AI outputs and fallback behavior while incorporating code-review feedback and reducing student-feedback turnaround 60% through rapid iterative releases.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Jumped into unfamiliar cross-platform code to **debug and optimize API components**, using structured hypotheses, SQL, caching, and response filtering to improve latency 30%, pipeline efficiency 68%, and maintainability.
- Collaborated with Microsoft and GitHub engineers through pull requests, code reviews, and feedback, automating versioned builds and 20+ unit and integration tests across Windows, Linux, and macOS releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **React and JavaScript web experiences** across eight Node.js services for authentication, records, permissions, notifications, and dashboards, synchronizing 2M+ records in real time through WebSockets and Azure SQL.
- Improved web performance and stability with browser testing, Sentry, Azure Monitor, and Kibana, reducing synchronization conflicts 33% and deployment time 50% while shipping reviewed, tested Docker releases through Jenkins.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built **responsive React and TypeScript interfaces** with HTML/CSS, real-time Firebase application state, and Flask APIs for bookings, pricing, payments, and status updates across 50,000+ records with sub-200-millisecond responses.
- Translated client needs into completed web features, mapping 12+ integrations and explaining technical decisions clearly while improving cross-platform compatibility 15%, reducing design errors 25%, and accelerating delivery 22%.

Skills

Languages: TypeScript, JavaScript, HTML, CSS, Python, SQL, Java
Web: React.js, Next.js, browser APIs, network fundamentals, async behavior, application state, WebSockets
UI: Responsive design, component libraries, React hooks, Shadcn, TailwindCSS, Material UI, dynamic content
AI: OpenAI, Claude, Qwen, LLM APIs, prompt chains, AI output validation, AI-assisted workflows
Quality: Jest, Selenium, PyTest, Git, GitHub, pull requests, code reviews, CI/CD, Sentry, performance optimization

Projects

Big-Brain Live | *React, TypeScript, Convex, LLM APIs*

- Built a **real-time collaborative learning application** with secure uploads, reusable React components, Convex synchronization, semantic search, and LLM summarization, processing 50,000+ documents while reducing retrieval latency 40% and review time 60%.

PiXMod Live | *Next.js, PyTorch, MongoDB*

- Created an **interactive web image application** for segmentation, recoloring, restoration, search, and subscription workflows, integrating SAM, Stable Diffusion, and MongoDB while processing 1,000+ edited assets through a responsive Next.js interface.

FinNexus Live | *Next.js, Kubernetes, MongoDB*

- Developed a **performant Next.js and TypeScript web application** supporting 50,000+ users and 1M+ transactions with responsive interfaces, real-time WebSocket updates, advanced filtering, secure access, and Kubernetes-based CI/CD automation.